

Assignment-3

1. Write a program to find the smallest and largest digit in a number.

```
In [4]: n = int(input("Enter a number: "))
smallest = 9
largest = 0
while n > 0:
    digit = n % 10

    if digit < smallest:
        smallest = digit

    if digit > largest:
        largest = digit

    n = n // 10

print("Smallest digit =", smallest)
print("Largest digit =", largest)
```

Enter a number: 569241

Smallest digit = 1

Largest digit = 9

2. Write a program to find the frequency of each digit in a number.

```
In [7]: n = int(input("Enter a number: "))
for i in range(10):
    count = 0
    temp = n
    while temp > 0:
        digit = temp % 10
        if digit == i:
            count = count + 1

        temp = temp // 10

    if count > 0:
        print(i, "=", count, "times")
```

Enter a number: 1223325555

1 = 1 times

2 = 3 times

3 = 2 times

5 = 4 times

3. Write a program to check whether a number is binary or not.

In [12]: `n = int(input("Enter a number: "))`

```
temp = n
binary = True

while temp > 0:
    digit = temp % 10

    if digit != 0 and digit != 1:
        binary = False
        break

    temp = temp // 10

if binary:
    print("Binary number")
else:
    print("Not a binary number")
```

Enter a number: 101

Binary number

4. Write a program to count how many digits are even and odd in a number.

In [14]: `n = int(input("Enter a number: "))`

```
even = 0
odd = 0
while n > 0:
    digit = n % 10

    if digit % 2 == 0:
        even = even + 1
    else:
        odd = odd + 1

    n = n // 10

print("Number of even digits =", even)
print("Number of odd digits =", odd)
```

Enter a number: 123456789

Number of even digits = 4

Number of odd digits = 5

5. Write a program to print the multiplication table of a given number.

```
In [16]: n = int(input("Enter a number: "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

Enter a number: 5

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

7. Write a Python program to construct the following pattern, using a nested for loop.

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
* *
```

```
In [20]: for i in range(1, 6):
for j in range(1, i + 1):
    print("*", end=" ")
print()
for i in range(4, 0, -1):
    for j in range(1, i + 1):
        print("*", end=" ")
    print()
```

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
* *
```

8. Print numbers from 1 to N, but skip numbers whose digit sum is divisible by 3.

```
In [1]: n = int(input("Enter N: "))

for i in range(1, n + 1):
    temp = i
    sum = 0

    while temp > 0:
        digit = temp % 10
        sum = sum + digit
        temp = temp // 10

    if sum % 3 == 0:
        continue
    else:
        print(i)
```

Enter N: 10

1
2
4
5
7
8
10

9.Count numbers between 1 and 100 whose reverse is greater than the original number.

```
In [26]: count = 0
for i in range(1, 101):
    temp = i
    reverse = 0

    while temp > 0:
        digit = temp % 10
        reverse = reverse * 10 + digit
        temp = temp // 10

    if reverse > i:
        count = count + 1

print("Count =", count)
```

Count = 36

10.Print all numbers between 1 and 100 where product of digits > sum of digits.

```
In [29]: for i in range(1, 101):
        temp = i
        sum = 0
        product = 1

        while temp > 0:
            digit = temp % 10

            sum = sum + digit
            product = product * digit

            temp = temp // 10

        if product > sum:
            print(i)
```

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99

11. Print all numbers from 1 to 100 that are not divisible by any digit they contain.

```
In [2]: for i in range(1, 101):  
        temp = i  
        divisible = False  
  
        while temp > 0:  
            digit = temp % 10  
            temp = temp // 10  
  
            if digit == 0:  
                continue  
  
            if i % digit == 0:  
                divisible = True  
                break  
  
        if divisible == False:  
            print(i)
```

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76
78
79
83
86
87
89
94
97
98

12. Print all pairs (i, j) from 1 to 10 such that:

☐ i + j is even

☐ i * j is divisible by 3

```
In [33]: for i in range(1, 11):
          for j in range(1, 11):

              if (i + j) % 2 == 0 and (i * j) % 3 == 0:
                  print("(", i, ",", j, ")")
```

```
( 1 , 3 )
( 1 , 9 )
( 2 , 6 )
( 3 , 1 )
( 3 , 3 )
( 3 , 5 )
( 3 , 7 )
( 3 , 9 )
( 4 , 6 )
( 5 , 3 )
( 5 , 9 )
( 6 , 2 )
( 6 , 4 )
( 6 , 6 )
( 6 , 8 )
( 6 , 10 )
( 7 , 3 )
( 7 , 9 )
( 8 , 6 )
( 9 , 1 )
( 9 , 3 )
( 9 , 5 )
( 9 , 7 )
( 9 , 9 )
( 10 , 6 )
```

13. Swap the first and last digit of a number.

```
In [35]: n = int(input("Enter a number: "))
temp = n
last = n % 10
div = 1

while temp >= 10:
    temp = temp // 10
    div = div * 10

first = temp
middle = (n % div) // 10
print("Number after swapping =", last * div + middle * 10 + first)
```

Enter a number: 12345

Number after swapping = 52341

14. Count how many numbers between 1 and 100 contain exactly two digits same.

```
In [3]: count = 0

for i in range(1, 101):
    temp = i
    digit_count = 0
    first_digit = -1

    while temp > 0:
        digit = temp % 10

        if first_digit == -1:
            first_digit = digit
        elif digit == first_digit:
            digit_count = digit_count + 1

        temp = temp // 10

    if digit_count == 1:
        count = count + 1

print("Count =", count)
```

Count = 10

15. Check whether a number is a Harshad number (divisible by sum of digits).

```
In [40]: n = int(input("Enter a number: "))

temp = n
sum = 0

while temp > 0:
    digit = temp % 10
    sum = sum + digit
    temp = temp // 10

if n % sum == 0:
    print("Harshad number")
else:
    print("Not a Harshad number")
```

Enter a number: 18

Harshad number

16. Find numbers whose square ends with the same digits as the number itself.(Hit: 5->5^2=25)

```
In [42]: for i in range(1, 101):  
        square = i**2  
        temp = i  
        divisor = 1  
  
        while temp > 0:  
            divisor = divisor * 10  
            temp = temp // 10  
  
        if square % divisor == i:  
            print(i)
```

```
1  
5  
6  
25  
76
```

17. Find numbers between 1 and 1000 that become prime after removing one digit.

```
In [5]: for n in range(10, 1001):

    temp = n
    digits = 0

    while temp > 0:
        digits = digits + 1
        temp = temp // 10

    divisor = 1
    for i in range(1, digits):
        divisor = divisor * 10

    temp = n
    found = False

    while divisor > 0:

        left = n // (divisor * 10)
        right = n % divisor

        new_number = left * divisor + right

        if new_number < 2:
            divisor = divisor // 10
            continue

        prime = True
        j = 2

        while j * j <= new_number:
            if new_number % j == 0:
                prime = False
                break
            j = j + 1

        if prime:
            found = True
            break

        divisor = divisor // 10

    if found:
        print(n)
```

39
42
43
45
47
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51
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62
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65
67
70



18. Print all numbers from 1 to 500 that are divisible by every non-zero digit they contain.

```
In [46]: for n in range(1, 501):

    temp = n
    valid = True

    while temp > 0:
        digit = temp % 10
        temp = temp // 10

        if digit == 0:
            continue

        if n % digit != 0:
            valid = False
            break

    if valid:
        print(n)
```

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400
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408
412
420
424
432
440
444
448
480
488
500

19. Print numbers from 1 to 100 but:

- ☒ Skip multiples of 4
- ☒ Stop completely if a number divisible by 17 appears

In [47]:

```
for i in range(1, 101):  
    if i % 17 == 0:  
        break  
  
    if i % 4 == 0:  
        continue  
  
    print(i)
```

1
2
3
5
6
7
9
10
11
13
14
15

20. Find the LCM and GCD of two numbers.

In [49]:

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
x = a  
y = b  
  
while y != 0:  
    remainder = x % y  
    x = y  
    y = remainder  
  
gcd = x  
lcm = (a * b) // gcd  
  
print("GCD =", gcd)  
print("LCM =", lcm)
```

Enter first number: 12
Enter second number: 18

GCD = 6
LCM = 36